

Panel MS-AR(1) for the log gap to the United States

Seasonally adjusted real GDP per worker

1 Model

The outcome is the log gap to the US in the same calendar quarter,

$$d_{it} = y_{it} - y_{US,t}, \quad (1)$$

where y is log seasonally adjusted real GDP per worker (period-average USD) from the 2026-09-17 panel. The United States is the numeraire and is dropped from the panel. A country-quarter is kept only if both that country and the US are observed. There is no common time trend: the US path already removes a shared gt . Joint panel Markov-switching AR(1) with three regimes:

$$d_{it} = a_i + z_{it}, \quad (2)$$

$$z_{i,t+1} = (1 - \rho(s_{it}))\mu(s_{it}) + \rho(s_{it})z_{it} + \sigma(s_{it})\varepsilon_{it}, \quad (3)$$

$$s_{i,t+1} \sim \Pi(\cdot | s_{it}), \quad (4)$$

$$a_i \sim \mathcal{N}(\alpha, \omega^2). \quad (5)$$

Spec 1 is this model. Spec 2 keeps the permanent gap and adds a decaying entry deviation from that gap,

$$d_{it} = a_i + b_i \lambda^{t-T_{i0}} + z_{it}, \quad (6)$$

$$b_i \sim \mathcal{N}(0, \omega_b^2) \text{ independent of } a_i, \quad (7)$$

with $\lambda = 0.98$ fixed (Barro 2% per year). T_{i0} is country i 's first observation. The long-run gap is a_i . Random effects are integrated by Gauss–Hermite quadrature (11-point in spec 1; 7×7 in spec 2). Plotted cycles use posterior-mean a_i (and b_i in spec 2). Latent paths s_{it} are country-specific. σ and ρ switch. The median μ is pinned at 0. $|\rho| < 0.99$. Standard errors (in parentheses) are delta-method from a numerical Hessian. π is the ergodic distribution of Π . $E[z]$ is the long-run mean of the cycle.

2 Log gap to the US: random intercepts, no trend

2026-09-17 SA panel, log gap to the US. 73 countries (USA dropped), 8364 observations, 1963-Q1–2026-Q2. Fitted countries: 73. Observations: 8364. Log-likelihood: 21706.11. Ergodic mean of the cycle $E[z] = 0.2281$ (0.0295). Random intercepts $a_i \sim N(\alpha, \omega^2)$ with $\alpha = -1.7591$ (0.0343), $\omega = 0.6689$ (0.0091). Posterior-mean a_i : mean -1.669, min -3.675, max 0.157.

Table 1: Regime AR(1), transition matrix, and ergodic π . Columns are regimes.

	(0)	(1)	(2)
μ	-0.2900 (0.1637)	0.0000 —	0.4435 (0.0372)
σ	0.0940 (0.0040)	0.0227 (0.0007)	0.0093 (0.0002)
ρ	0.9900 (0.0001)	0.9900 (0.0000)	0.9900 (0.0000)
Π_0	0.5808 (0.0440)	0.4104 (0.0509)	0.0088 (0.0201)
Π_1	0.0563 (0.0090)	0.8770 (0.0134)	0.0666 (0.0092)
Π_2	0.0133 (0.0032)	0.0323 (0.0060)	0.9544 (0.0051)
π	0.0679 (0.0076)	0.3733 (0.0236)	0.5588 (0.0267)

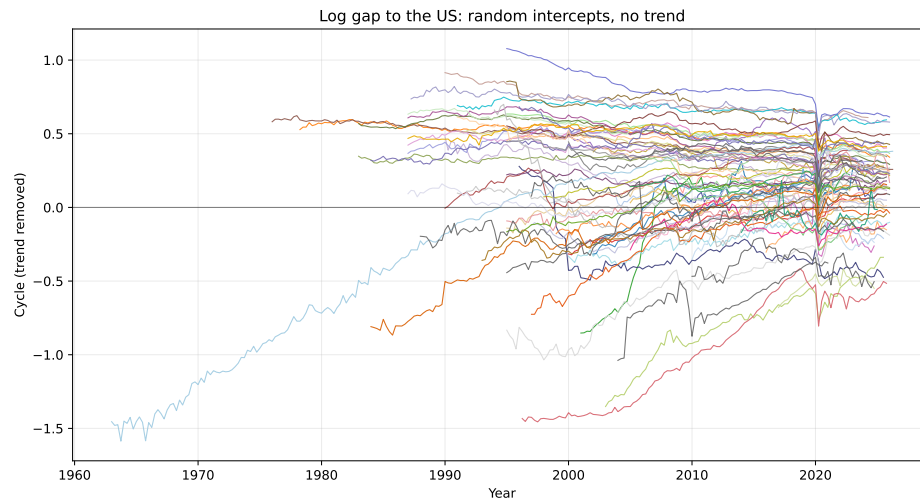


Figure 1: Country cycles in the US gap after removing the deterministic path.

3 Log gap to the US: RE intercepts plus Barro catch-up

2026-09-17 SA panel, log gap to the US. 73 countries (USA dropped), 8364 observations, 1963-Q1–2026-Q2. Fitted countries: 73. Observations: 8364. Log-likelihood: 21810.29. Ergodic mean of the cycle $E[z] = 0.0284$ (0.0409). Permanent RE intercepts and catch-up $y_{it} = a_i + b_i \lambda^{t-T_{i0}} + z_{it}$ with $\alpha = -1.4034$ (0.0409), $\omega = 0.6483$ (0.0080), $\lambda = 0.9800$ (fixed), $\omega_b = 0.4701$ (0.0123). Posterior-mean a_i : mean -1.495, min -3.834, max 0.132. Posterior-mean b_i : mean -0.023, min -1.106, max 1.113.

Table 2: Regime AR(1), transition matrix, and ergodic π . Columns are regimes.

	(0)	(1)	(2)
μ	-0.9403 (0.1739)	0.0000 —	0.2559 (0.0408)
σ	0.0910 (0.0039)	0.0217 (0.0007)	0.0089 (0.0002)
ρ	0.9830 (0.0052)	0.9900 (0.0000)	0.9900 (0.0000)
Π_0	0.5877 (0.0420)	0.4123 (0.0420)	0.0000 (0.0000)
Π_1	0.0544 (0.0077)	0.8826 (0.0107)	0.0630 (0.0076)
Π_2	0.0126 (0.0030)	0.0338 (0.0055)	0.9536 (0.0048)
π	0.0685 (0.0077)	0.3950 (0.0238)	0.5365 (0.0271)

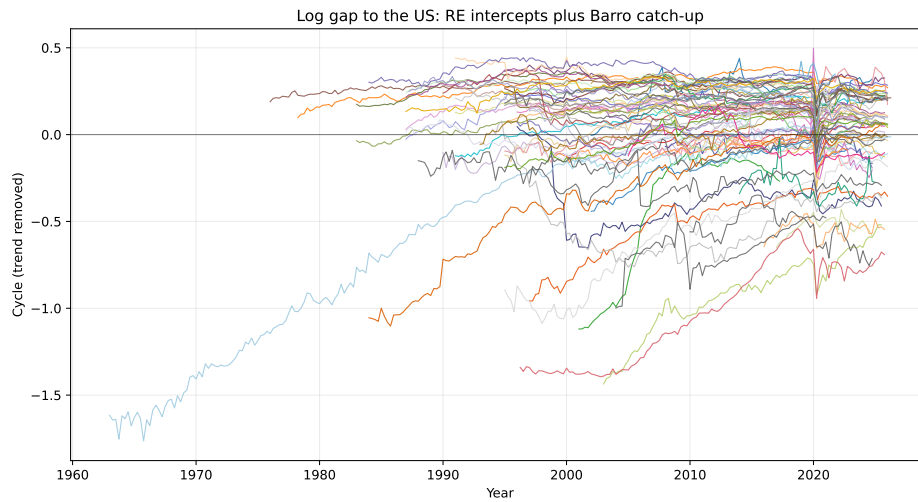


Figure 2: Country cycles in the US gap after removing the deterministic path.